

Developer Setup Guide

This is a simple guide to set up the repositories for the CSV processor application code, infrastructure configs and manifests.

Repository Structure

```
workspace/  
■■■ csv-processor-app/  
■■■ csv-processor-infrastructure/  
■■■ csv-processor-k8s-assets/  
  
git clone https://github.com/asif-ahmedb/csv-processor-app  
git clone https://github.com/asif-ahmedb/csv-processor-infrastructure  
git clone https://github.com/asif-ahmedb/csv-processor-k8s-assets
```

Prerequisites

```
All Paths:  
- Git 2.x  
- Docker Desktop 4.x  
  
Local / Minikube:  
- Minikube  
- kubectl  
- Helm 3  
- Python 3.10+  
  
AWS:  
- AWS CLI v2  
- kubectl  
- Helm 3  
- Git Bash  
- Kops (for Kops deployments)  
- Terraform 1.5+ (for EKS deployments)
```

AWS IAM Permissions

```
Required services:  
EC2, VPC, ELB, S3, IAM, Route53 (Kops), EKS (Terraform)  
  
Validation:  
aws sts get-caller-identity
```

Path 1 – Local Deployment

```
cd csv-processor-app  
docker build -t csv-processor:local .  
  
docker run --rm -p 8081:8081 -e S3_BUCKET="" -e AWS_REGION=us-east-1 csv-processor:local  
  
Open:  
http://localhost:8081  
  
Run tests:  
cd csv-processor-app/web  
pip install -r requirements.txt  
pytest tests/ -v  
  
Minikube:  
cd csv-processor-k8s-assets/minikube  
./deploy.sh  
  
kubectl port-forward -n csv-processor svc/csv-processor 8080:80
```

Path 2 – AWS with Kops

```
cd csv-processor-infrastructure/kops
cp settings.env.example settings.env

Set:
AWS_ACCOUNT_ID=<ACCOUNT_ID>
AWS_REGION=us-east-1

chmod +x scripts/*.sh
./scripts/bootstrap.sh

Deploy:
helm upgrade --install csv-processor ...

Access:
kubectl get svc -n csv-processor csv-processor -w

Teardown:
./scripts/teardown.sh --yes
```

Path 3 – AWS with Terraform/EKS

```
cd csv-processor-infrastructure/terraform
terraform init
terraform plan
terraform apply

aws eks update-kubeconfig ...

Deploy:
helm upgrade --install csv-processor ...

Access:
kubectl port-forward -n csv-processor svc/csv-processor 8080:80

Teardown:
terraform destroy
```

CI/CD

```
GitHub Actions workflow:
- Run pytest
- Build Docker image
- Push Docker image to Docker Hub

Required secrets:
DOCKERHUB_USERNAME
DOCKERHUB_TOKEN
```

Reference Documents

```
Kops:
https://github.com/asif-ahmedb/csv-processor-infrastructure/blob/main/kops/KOPS\_QUICK\_REFERENCE.md

Deployment:
https://github.com/asif-ahmedb/csv-processor-k8s-assets/blob/main/DEPLOY\_QUICK\_REFERENCE.md

Minikube:
https://github.com/asif-ahmedb/csv-processor-k8s-assets/blob/main/minikube/MINIKUBE\_QUICK\_REFERENCE.md
```